

Multi-field Balance-aware Calibration of Predictions in Online Advertising

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Abstract

Online advertising platforms serve as a critical bridge between advertisers and media, requiring precise prediction of user behaviors. Systematic underestimation or overestimation in these predictions can undermine the interests of both parties. Existing calibration methods fall short in addressing two key challenges. First, significant differences in CVR distributions across various targets lead to biased calibration when using global posterior statistics, causing some sample groups to be overestimated while others are underestimated. Second, current field-aware approaches are typically limited to single-field calibration and fail to account for field sensitivity. To overcome these limitations, we propose a Pareto Frontier-based Multi-field Personalized Calibration (PF-MPC) method which formulates multi-field calibration as a multi-objective optimization problem. PF-MPC identifies the optimal Pareto-efficient weight combinations to balance the conflicting calibration errors across different fields. We evaluate PF-MPC on both public calibration benchmark and a large-scale industrial dataset. Experimental results demonstrate that our method achieves significant improvements in calibration performance compared to existing approaches.

CCS Concepts

• Information systems → Computational advertising.

Keywords

Multi-field Calibration, Uncertainty calibration, Multi-objective Optimization, Online Advertising

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1 Introduction

In the ranking stage of online advertising systems, it is critical to ensure not only the relative ordering accuracy of items but also the numerical reliability of the predicted probabilities [2, 4]. Under the application of optimized Cost-Per-X (oCPX) advertising, the predicted click-through rate (CTR) and conversion rate (CVR) for each impression directly determine accuracy and stability of bidding. From the platform's perspective, prediction accuracy further influences both advertiser settlements and platform revenue. Specifically, consistent overestimation prevents advertisers from acquiring the expected volume of valuable traffic, undermining their trust in the platform. Conversely, persistent underestimation leads to undervalued transactions and reduced platform and media earnings.

Model calibration is a classic machine learning task that eliminates the discrepancy between predicted probabilities and true probabilities [5, 13]. As a post-processing step in online advertising, it corrects the output of the ranking model by leveraging collected posterior data [2]. This process ensures that the predicted values constitute a faithful reflection of the observed empirical probabilities. The resulting calibrated estimates are then employed in subsequent ranking and bidding stages.

However, classical calibration methods such as Platt Scaling [8] and Histogram Binning [12] are often inadequate for real-world advertising systems. This is primarily due to their reliance on strong assumptions and failure to model complex, non-stationary online distributions. More recent field-aware calibration methods, such as NeuralCalib [7], AdaCalib [10], and DESC [11], aim to address distribution shifts by incorporating domain information into the calibration process. These approaches typically employ neural networks to adaptively learn the parameters of transformation functions, thereby tailoring the calibration for samples with different feature characteristics.

Despite these advancements, two fundamental limitations persist. First, these deep methods still rely heavily on global posterior statistics. While the overall calibration metrics may be optimized, systematic biases emerge at the local sample level: predictions

for certain sample groups are consistently overestimated, whereas those for others are persistently underestimated. Second, although existing single-field calibration methods account for distributional differences within the calibration space of a specific field, they overlook potential interactions across multiple features. Consequently, improvements in calibration performance for one feature often come at the expense of degraded performance for others—a phenomenon directly reflected in the deterioration of field-specific metrics such as Field-Level Expected Calibration Error (F-ECE). For instance, even if we achieve robust calibration in the ad domain, inadequate calibration may persist in the media domain.

To address these limitations, we propose a Pareto Frontier-based Multi-field Personalized Calibration (PF-MPC) method, which formulates multi-field calibration as a multi-objective optimization problem. Our approach consists of two core components. First, PF-MPC employs multiple base domain-specific calibrators, each designed to be isotonic. Guided by the Unconstrained Monotonic Neural Network (UMNN) architecture[1, 9], each calibrator takes both the raw prediction score and field-specific features as input, and is optimized to minimize the Expected Calibration Error (ECE) for its corresponding field, thereby ensuring field-level calibration optimality. Second, to harmonize calibration across multiple fields, PF-MPC leverages Pareto Frontier to identify Pareto-efficient weight combinations that balance conflicting ECE objectives across fields. Additionally, our method supports end-to-end training, eliminating the need for independent per-field calibrator optimization. The main contributions of this paper are summarized as follows:

- We propose the PF-MPC framework for multi-field level calibration. The domain-specific calibrator within the framework is designed to satisfy the isotonicity constraint, which is implemented using an Unconstrained Monotonic Neural Network.
- A Pareto Frontier-based mechanism is designed to ensure that the learned joint calibration achieves balance across multiple fields.
- Extensive experiments confirm that PF-MPC achieves superior calibration performance over existing SOTA methods.

2 Preliminaries

2.1 Problem Definition

Consider a pre-trained neural fine-ranking model $f_{\text{uncalib}}(\cdot)$ trained on a training dataset $\mathcal{D}_{\text{train}} = \{(x, y)\}$ where x represents a sample's feature information (encompassing user fields, item fields, media fields, and context fields) and $y \in \{0, 1\}$ denotes the sample's binary response label (e.g., conversion). The model outputs an uncalibrated prediction:

$$\hat{p} = f_{\text{uncalib}}(x) \quad (1)$$

Here, $\hat{p} \in (0, 1)$ is the raw probability score generated by the ranking model. However, $f_{\text{uncalib}}(\cdot)$ often fail to align with the true underlying likelihood of the binary response because the true sample-wise likelihood is not directly observable in practice.

To address this prediction discrepancy, we learn a calibrator f_{calib} using a validation dataset with n distinct fields $Z_1, Z_2, \dots, Z_n$: its function is to post-process the uncalibrated score $\hat{p}$ to a calibrated probability $\hat{p}_{\text{calib}}$.

2.2 Calibration Metrics

Field-level calibration errors measure prediction accuracy across different dataset subsets. A categorical feature $z \in \mathcal{Z}$ is chosen as the partition field, splitting the dataset into $\|\mathcal{Z}\|$ disjoint subsets. The Field-level Expected Calibration Error (F-ECE) is defined as a weighted sum of the average prediction bias in each subset

$$F\text{-ECE} = \frac{1}{\mathcal{D}} \sum_{z=1}^{|\mathcal{Z}|} \left| \sum_{i=1}^{|\mathcal{D}|} (y_i - \hat{p}_i) \mathbf{1}_{[z_i=z]} \right| \quad (2)$$

The field-level relative calibration error formulated as the averaged rate of errors divided by the true label,

$$F\text{-RCE} = \frac{1}{\mathcal{D}} \sum_{z=1}^{|\mathcal{Z}|} N_z \frac{\left| \sum_{i=1}^{|\mathcal{D}|} (y_i - \hat{p}_i) \mathbf{1}_{z_i=z} \right|}{\left| \sum_{i=1}^{|\mathcal{D}|} (y_i + \epsilon) \mathbf{1}_{z_i=z} \right|} \quad (3)$$

where N_z is the number of instance in subsets, i.e., $\sum_{z=1}^{|\mathcal{Z}|} |\mathcal{D}| = |\mathcal{D}|$.

F-ECE and F-RCE can only measure the calibration performance of a single field. To evaluate the calibration error across multiple fields, Multi-Field ECE (MF-ECE) and Multi-Field RCE (MF-RCE) are required, with their formulations provided as follows:

$$MF\text{-ECE} = \frac{1}{n} \sum_i F\text{-ECE}_i \quad (4)$$

where n is the number of fields.

3 Framework

In this section, we present our proposed calibration framework for multi-field machine learning models, particularly tailored for recommendation systems involving fields such as ad, user and media features.

3.1 Intra-Field Calibration

To achieve precise calibration within fields, we first group all fields into several categories based on offline data, such as item features and user features. This grouping results in n independent calibrators, each corresponding to a specific subset of features.

The calibration logic for each calibrator is as follows: It takes the raw logits output from the base model as input and combines them with the allocated feature subset to construct contextual information. The adjustment to the raw logits is performed via Equation (5), yielding the calibrated logits under this feature perspective:

$$g(f(x)) = \int_0^{f(x)} f(t)dt + h(x) \quad (5)$$

where $g(\cdot)$ represents the overall calibration transformation function ensuring monotonic alignment; $f(x)$ denotes the processed input combining logits and features; $h(\cdot)$ is an auxiliary density function implemented by a multi-layer perceptron (MLP) network, serving as a bias integrator for the current feature group to maintain monotonicity and calibration precision; and the integral form enforces strict monotonicity in the mapping, with the output representing the calibrated result from a single perspective.

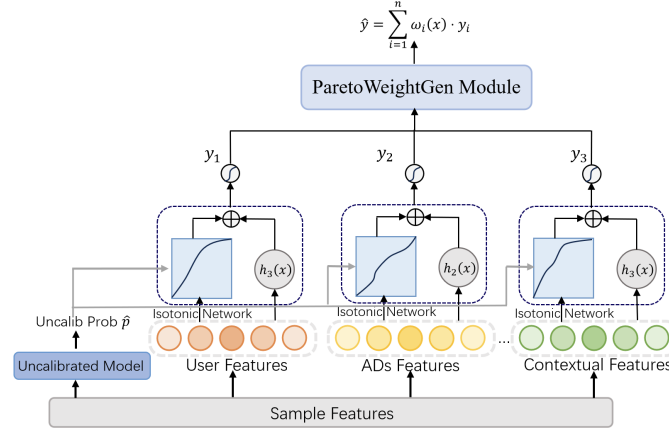


Figure 1: Overview of the proposed multi-field calibration framework. The uncalibrated model processes input features to generate uncalibrated probabilities $\hat{p}$. These are then calibrated intra-field using Isotonic Networks for each field group, producing calibrated outputs $y_1, y_2, y_3, \dots$. Finally, the ParetoWeightGen module, generates dynamic normalized weights $w_i(x)$ to fuse the outputs via a weighted sum $\sum w_i(x) \cdot y_i$, achieving balanced inter-field calibration through Pareto optimization.

This process ensures that predictions within each grouped field are reliably aligned with true probabilities, leveraging monotonic networks [1], field-specific features, and posterior probabilities.

3.2 Inter-Field Calibration

After intra-field calibration, we integrate all calibrated outputs from multiple fields using a learnable multi-layer perceptron (MLP) module, denoted as ParetoWeightGen. This module generates dynamic, normalized fusion weights that adapt to each sample's global context, achieving balanced calibration through Pareto optimization.

Formally, for each input sample x , let the calibrated outputs from n fields be $[y_1, y_2, \dots, y_n]$. The fusion module first predicts a set of unnormalized scores $[\hat{\omega}_1, \hat{\omega}_2, \dots, \hat{\omega}_n]$ via an MLP network. To ensure stability and interpretability, the scores are normalized using a Softmax function:

$$\omega_i(x) = \frac{\exp(\hat{\omega}_i)}{\sum_{j=1}^n \exp(\hat{\omega}_j)}, \quad \text{s.t.} \quad \sum_{i=1}^n \omega_i(x) = 1$$

The final calibrated prediction is then obtained as a weighted sum of field-specific outputs:

$$\hat{y} = \sum_{i=1}^n \omega_i(x) \cdot y_i$$

3.3 Field-Level Calibration Loss

We divide the fields into several field groups (indexed by k) to reduce the number of Pareto objectives. For each field group k containing the field set S_k , the group calibration loss S_k is defined as the average of Field-wise Relative Calibration Errors (FRCE) of all fields in the group:

$$L_k = \frac{1}{|S_k|} \sum_{i \in S_k} FRCE_i$$

Here, B is the current batch, M equal-frequency bins G_b^i are constructed based on the calibrated confidence $\hat{y}_i$ of field i , $\bar{y}_b^i$ is the empirical accuracy (average true label) in bin b for field i , and $\epsilon = 1e^{-6}$ prevents division by zero. This relative formulation penalizes miscalibration more heavily in low-click-rate bins, which is particularly valuable in highly imbalanced CTR tasks.

3.3.1 Pareto Loss Optimization. To achieve balanced multi-field calibration, we combine field-level losses L_k into a Pareto-optimized objective. For each field, we compute the gradient of L_k with respect to its calibrated output and aggregate them using the normalized weights ω_k from ParetoWeightGen. The Pareto loss is defined as:

$$L_{\text{Pareto}} = \left\| \sum_{k=1}^n \omega_k \cdot \nabla L_k \right\|^2$$

This term aligns the update directions of different fields, forcing the optimization to seek a Pareto-stationary solution [3] where no field's calibration performance can be improved without slightly degrading another. During training, only the parameters of the ParetoWeightGen network are updated, while calibrator and base model parameters remain frozen.

To further enhance convergence and balance, the optimization of multiple field group calibrators and the Pareto optimization are performed alternately within each batch, with separate optimizers. This alternating and separate optimization strategy decouples local field calibration from global fusion, reducing gradient conflicts and improving overall calibration consistency in multi-field scenarios.

4 Experiments

4.1 Experimental Settings

4.1.1 datasets. We evaluate our method on two datasets: the public AliCCP dataset for CTR prediction [6] and a large-scale industrial oCPX conversion dataset. The AliCCP dataset contains approximately 80 million samples, split 2:1:1 into training, validation,

Table 1: Performance comparison on calibration metrics (lower is better) and AUC (higher is better). All values are reported on the test set of the AliCCP dataset and industrial OCPX conversion dataset.

Method	CTR Task(public data)						CVR Task(industrial data)					
	AUC	PCOC	ECE	MFECE	FRCE	MFRCE	AUC	PCOC	ECE	MFECE	FRCE	MFRCE
Uncalib	0.6348	1.6097	0.0202	0.0230	1.4869	0.7327	0.9312	1.1615	0.2133	0.4351	0.3673	0.3341
HistoBin	0.5241	1.2282	0.0076	0.0118	1.2110	0.3909	0.8712	0.9573	0.1190	0.3276	0.2950	0.2212
IsoReg	0.6348	1.2352	0.0078	0.0117	1.0170	0.3557	0.9312	1.1214	0.1115	0.3465	0.2841	0.2123
PlattScaling	0.6348	1.2355	0.0078	0.0117	1.0180	0.3561	0.9252	1.1145	0.0823	0.3414	0.2947	0.2134
SIR	0.6347	1.2457	0.0083	0.0120	1.0869	0.3776	0.9312	1.1023	0.0651	0.3314	0.2812	0.1912
NeuralCalib	0.6357	1.2094	0.0070	0.0109	0.9906	0.3316	0.9342	1.0512	0.0368	0.1031	0.1434	0.1334
DESC	0.6352	1.2187	0.0073	0.0112	1.0018	0.3404	0.9336	1.0410	0.0312	0.0814	0.1528	0.0841
UMC	0.6358	1.1711	0.0057	0.0098	0.9496	0.2942	0.9348	1.0501	0.0212	0.1041	0.1423	0.1134
PF-MPC	0.6349	1.1032	0.0051	0.0084	0.8648	0.2270	0.9303	1.0261	0.0089	0.0312	0.1237	0.0613

and test sets. The industrial dataset comprises over 400 million anonymized samples from 7 days of advertising logs.

4.1.2 baseline. We compare our proposed method with a range of baselines to demonstrate its superiority in multi-field calibration. The uncalibrated baseline directly uses the raw prediction scores from the base model without any calibration. Histogram Binning [12] works by grouping predictions into bins and assigning the empirical accuracy of each bin as the calibrated probability; Platt Scaling [8] is a parametric approach that assumes a logistic distribution; SIR [2] introduces a linear interpolation strategy on top of IR to enhance calibration smoothness; NeuralCalib [7] combines a simple piece-wise linear model with a field-aware auxiliary neural network; DESC [11] leverages deep ensembles for scaling; and UMC [1] utilizes monotonic networks to learn infinitely smooth order-preserving functions for calibration.

4.1.3 Implementation Details. All calibration methods (including ours) are trained for exactly 1 epoch on the training set, with the base model completely frozen. Hyper-parameters are tuned on the development set and reported on the test set. For FRCE computation, the distinctive field is designated as '101' in the AliCCP dataset, consistent with prior methods for fair comparison.

4.2 Performance Comparison

As shown in Table 1, our PF-MPC method achieves state-of-the-art calibration performance on both the public AliCCP and industrial datasets. It significantly outperforms all baselines on field-aware metrics (MF-ECE, MF-RCE) and also leads on classic calibration metrics (ECE, PCOC). However, as its primary focus is calibration, it achieves competitive but not leading AUC.

5 Conclusion

This paper proposes a multi-field calibration framework named PF-MPC for calibrating online advertising predictions. The method leverages basic isotonic calibrators for intra-field calibration and Pareto front optimization to ensure balanced performance across defined domains. Empirical evaluation on two industrial datasets demonstrated that PF-MPC achieves state-of-the-art calibration

performance. Future work will focus on enhancing its isotonicity-preserving capability and extending the framework to online environment.

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